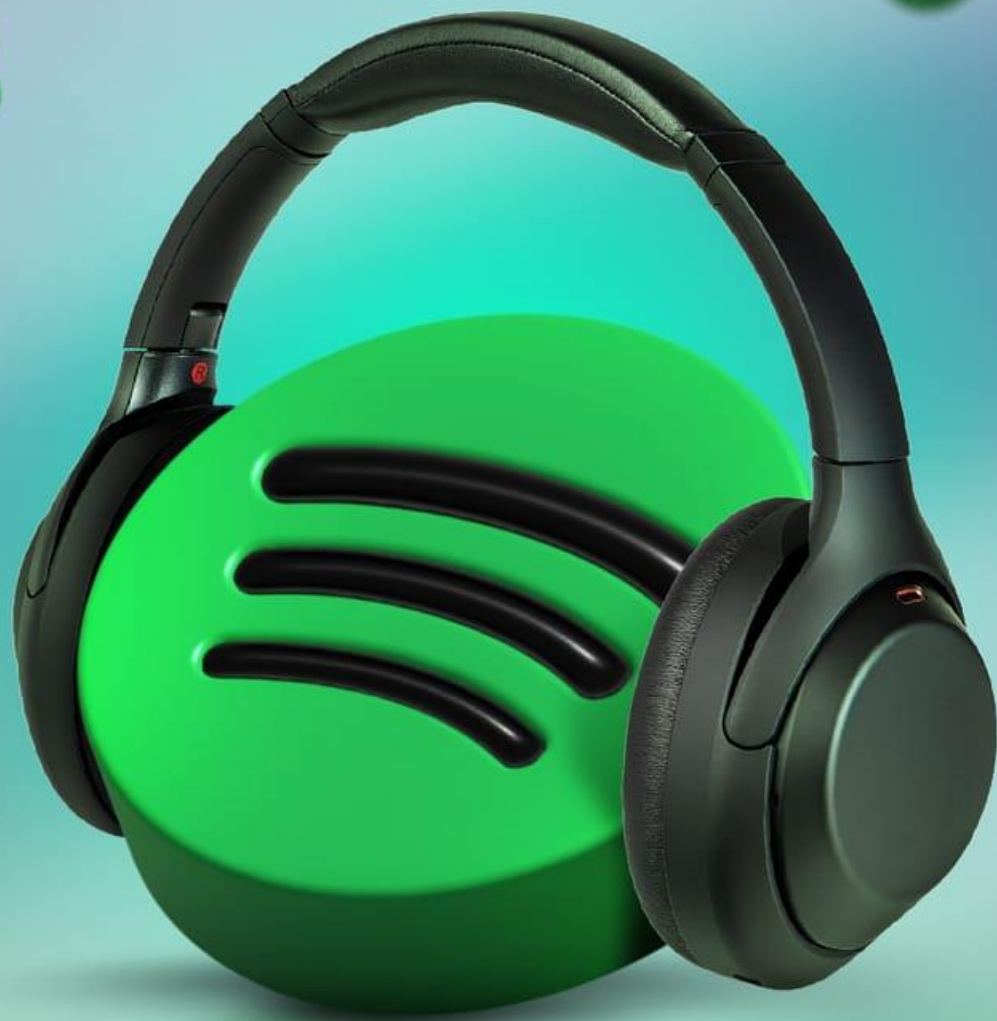






Spotify

MACHINE LEARNING



SALES GROWTH

€4,2 BILLION

Introduction

- Have you ever noticed how Spotify plays the perfect song at the perfect time? This is possible because Spotify uses AI and Machine Learning to understand user taste and behavior.
- Spotify is not just a music app. It is an AI-driven platform that continuously learns from users and music data to deliver a highly personalized experience.
- “Spotify doesn’t just play music—it understands you. And that intelligence comes from AI and Machine Learning.”

Recommendation System (Core of Spotify AI)

- What AI Does

Spotify uses Machine Learning recommendation algorithms to decide:

What song to play next

What playlist to suggest

What new artist you may like

- How It Works (Step-by-Step)

AI collects user data:

Songs played

Songs skipped

Likes & shares

Play duration

ML models analyze patterns

Spotify compares your taste with millions of users

System predicts songs you are most likely to enjoy

- AI Techniques Used

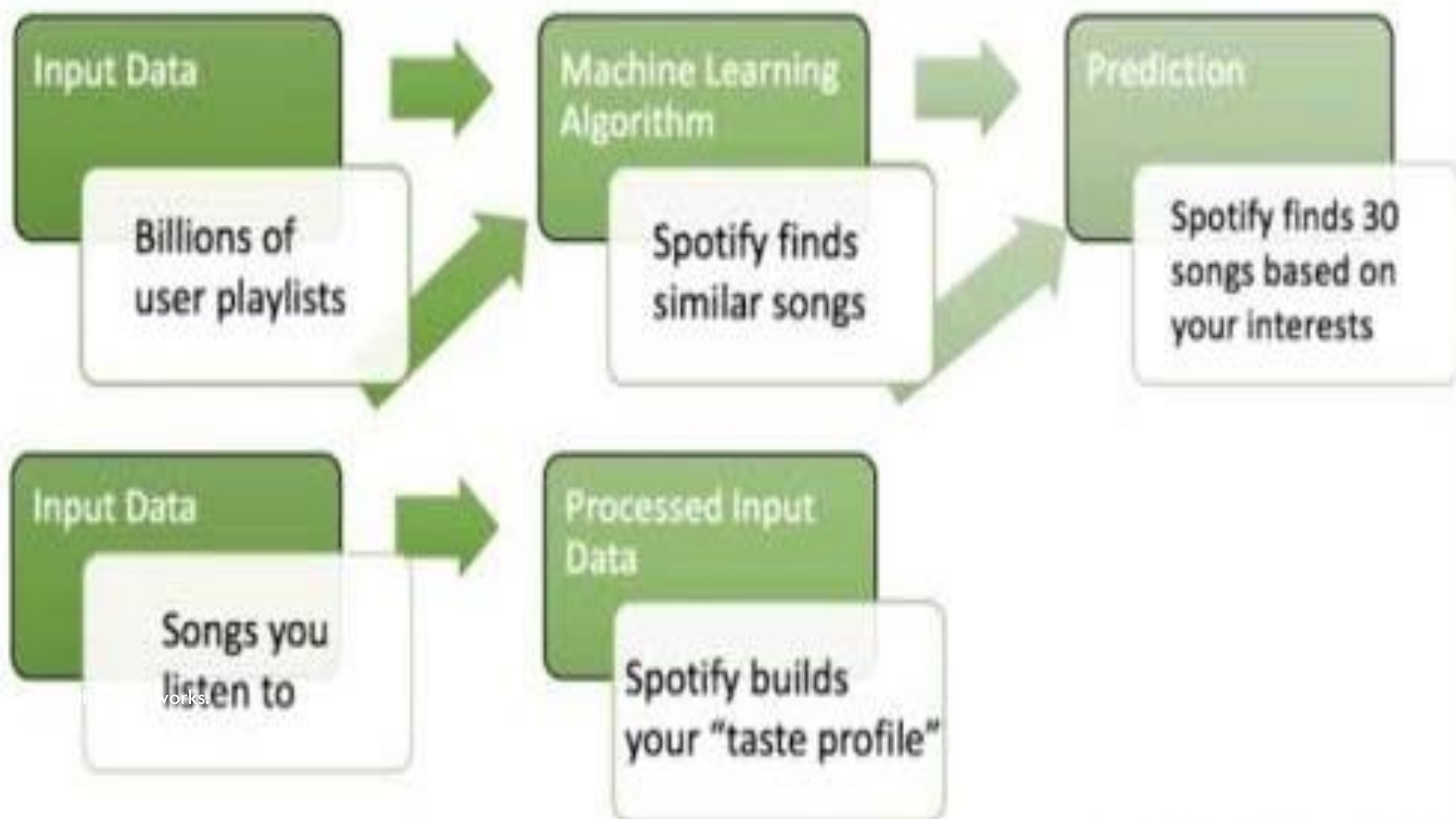
Collaborative Filtering

Neural Networks

Matrix Factorization

✚ Example:

If users with similar taste like Song A and Song B, Spotify recommends Song B to you.



Audio Content Analysis (Understanding Music Itself)

- Spotify does not only analyze user behavior; it also **analyzes the music itself using AI**.
- AI studies the **sound properties of songs** to understand how a song actually feels.
- **Audio Features Analyzed**
Spotify analyzes:
Tempo (speed)
Pitch
Loudness

Rhythm

Energy

Instrumentalness

Danceability

- **ML Role**

Converts audio into numerical data

Groups similar songs

Helps in genre classification automatically.

📌 Example:

Two songs with similar tempo and energy are grouped into **workout playlists**.

Natural Language Processing (NLP)

- Spotify uses **Natural Language Processing (NLP)** to understand **human language** given by users through **text input or voice commands**.
- NLP helps Spotify understand **what the user is asking**, not just the words typed.

- **How AI & ML Work in This Module**

When a user gives a command like:

✚ “**Play English songs**”

First, **AI receives the input.**

If the input is a voice command, AI converts **speech into text** using speech recognition.

Next, **NLP analyzes the text** and breaks the sentence into meaningful parts:

- **English** → identifies the language
- **Songs** → identifies the content type

This process is called **intent understanding**, where AI finds out *what the user wants*.

- **Role of Machine Learning**

After understanding the user command, **Machine Learning models:**
Analyze the user's **listening history**
Check **previous searches and liked songs**
Compare preferences with **similar users**
Based on this data, ML predicts the **most suitable songs or playlists.**

- **Final Output**

Spotify then:
Plays **relevant English songs**
Or recommends **matching playlists**
This makes the results **accurate, fast, and personalized.**

Personalized Playlists Creation

- ▶ AI Automatically Creates:
Discover Weekly
Daily Mix
Release Radar
Mood-based playlists
- ▶ ML Decision Factors
Time of day
Day of week
Listening history
Mood patterns 
- ▶ Example:
- ▶ Morning → soft / motivational
- ▶ songs
Night → calm / lofi music

Spotify uses AI to analyze songs

- ▶ Beat
- ▶ Pitch
- ▶ Rhythm
- ▶ Loudness
- ▶ Danceability
- ▶ This helps match songs with your mood or activity (gym, study, sleep).

Podcast & Audiobook Recommendation AI suggests:

- ▶ Podcast topics you like Episode order Audiobooks based on interests.

- ▶ ML suggests:

Podcasts you may like Episodes similar to what you listen to New creators matching your interest

- ▶ ✂ Example: If you listen to technology podcasts, Spotify suggests AI or startup podcasts.

Thank You